

Switching

• In large networks, there may be more than one path for transmitting data from **sender** to receiver. Selecting a path that data must take out of the available options is called **switching** (finding the best route to transfer data from sender to receiver). There are two popular switching techniques – circuit switching and packet switching.

- Switching in a computer network is achieved by using switches. A switch is a small hardware device which is used to join multiple computers together with one local area network (LAN).

- Network switches operate at layer 2 (Data link layer) in the OSI model.

- Switching is transparent to the user and does not require any configuration in the home network.

- Switches are used to forward the packets based on MAC addresses.

- A Switch is used to transfer the data only to the device that has been addressed. It verifies the destination address to route the packet appropriately.

It is operated in full duplex mode.

- Packet collision is minimum as it directly communicates between source and destination.

It does not broadcast the message as it works with limited bandwidth.

Why is Switching Concept required?

Switching concept is developed because of the following reasons:

Bandwidth: It is defined as the maximum transfer rate of a cable. It is a very critical and expensive resource. Therefore, switching techniques are used for the effective utilization of the bandwidth of a network.

Collision: Collision is the effect that occurs when more than one device transmits the message over the same physical media, and they collide with each other. To overcome this problem, switching technology is implemented so that packets do not collide with each other.



Advantages of Switching:

Switch increases the bandwidth of the network.

It reduces the workload on individual PCs as it sends the information to only that device which has been addressed.

It increases the overall performance of the network by reducing the traffic on the network.

There will be less frame collision as switch creates the collision domain for each connection.

Disadvantages of Switching:

A Switch is more expensive than network bridges.

A Switch cannot determine the network connectivity issues easily.

Proper designing and configuration of the switch are required to handle multicast packets.

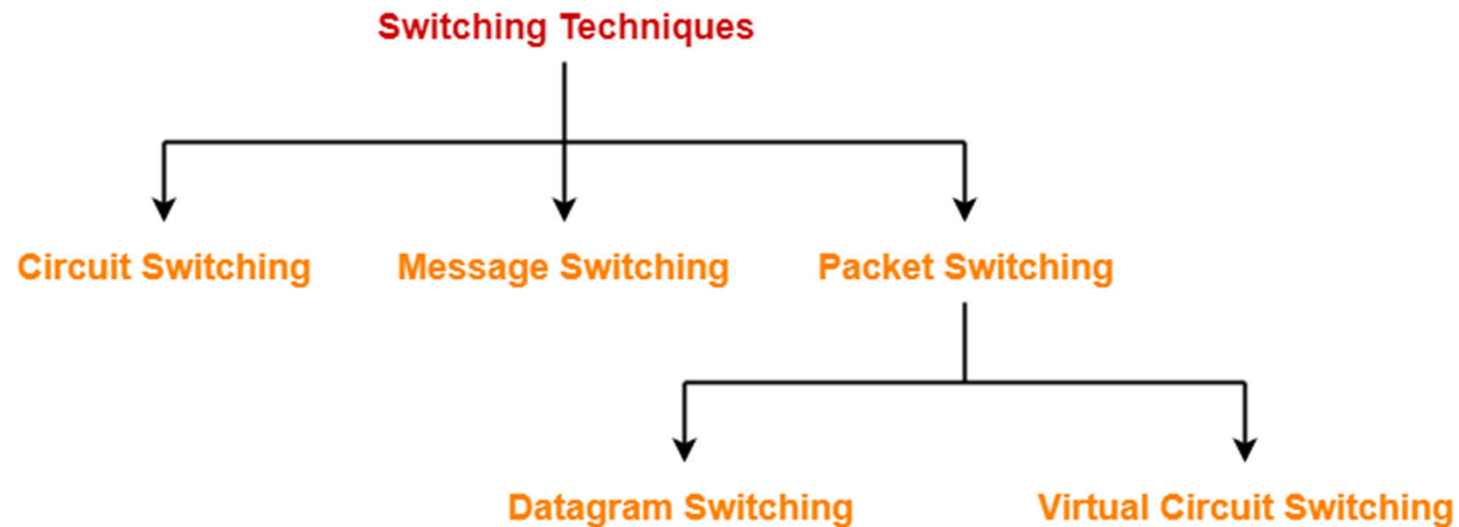


Switching techniques

In large networks, there can be multiple paths from sender to receiver. The switching technique will decide the best route for data transmission.

Switching technique is used to connect the systems for making one-to-one communication.

Classification Of Switching Techniques



Circuit Switching

Circuit switching is a switching technique that establishes/creates a dedicated path between sender and receiver. In the Circuit Switching Technique, once the connection is established then the dedicated path will remain to exist until the connection is terminated.

Circuit switching in a network operates in a similar way as the telephone works.

A complete end-to-end path must exist before the communication takes place.

In case of circuit switching technique, when any user wants to send the data, voice, video, a request signal is sent to the receiver then the receiver sends back the acknowledgment to ensure the availability of the dedicated path. After receiving the acknowledgment, dedicated path transfers the data.

Circuit switching is used in public telephone network. It is used for voice transmission (telephone network).

Fixed data can be transferred at a time in circuit switching technology.

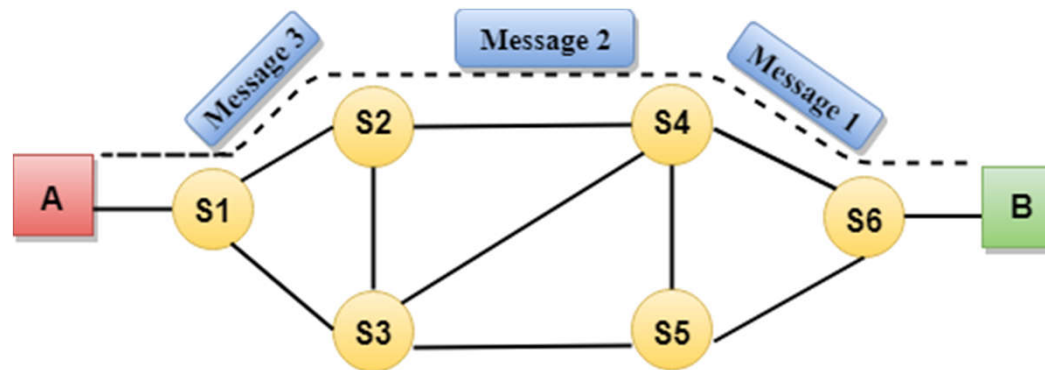
In circuit switching, each data unit know the entire path address which is provided by the source.

Communication through circuit switching has 3 phases:

Circuit establishment

Data transfer

Circuit Disconnect



Message Switching

Message Switching is a switching technique in which a message is transferred as a complete unit (divided into many packets) and routed through intermediate nodes at which it is stored and forwarded.

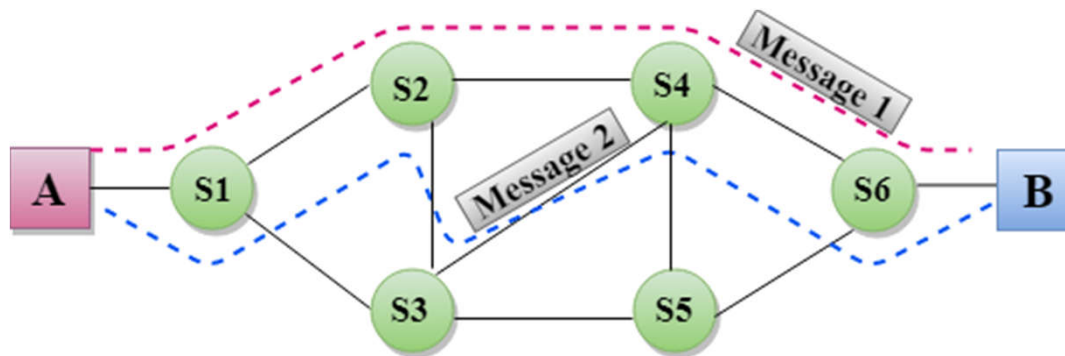
In Message Switching technique, there is no establishment of a dedicated path between the sender and receiver.

The destination address is appended to the message. Message Switching provides a dynamic routing as the message is routed through the intermediate nodes based on the information available in the message.

Message switches are programmed in such a way so that they can provide the most efficient routes.

Each and every node stores the entire message and then forward it to the next node. This type of network is known as **store and forward network**.

Message switching treats each message as an independent entity.



Packet switching is most commonly seen on **data networks, including LANs, WANs and the Internet**. It is well suited to data transmission, as it allows networks to adapt to changing conditions quickly. For example, if one of the network's routers fails, packets can be automatically re-routed to avoid that device.

Data packets are able to find the destination without the use of a dedicated channel. Reduces lost data packets because packet switching allows for resending of packets. More cost-effective since there is no need for a dedicated channel for voice or data traffic.

Packet switching is currently the most modern form of telephone system. Instead of using one line and one communication channel dedicated to a single call, packet switching divides the call data up into small units, known as packets.

Packet switching is easier and more affordable than circuit switching. Since all the bandwidth can be used at once, packet switching is more efficient because it doesn't have to deal with a limited number of connections that may not be using all that bandwidth. Voice-over-Internet protocol (VoIP) is **a technology that lets users make calls using a broadband Internet connection instead of a standard phone line**. VoIP technology converts the voice signal used in traditional phone calls into a digital signal that travels via the Internet rather than analog phone lines.

Advantages Of Message Switching

Data channels are shared among the communicating devices that improve the efficiency of using available bandwidth.

Traffic congestion can be reduced because the message is temporarily stored in the nodes.

Message priority can be used to manage the network.

The size of the message which is sent over the network can be varied. Therefore, it supports the data of unlimited size.

Disadvantages Of Message Switching

The message switches must be equipped with sufficient storage(secondary storage) to enable them to store the messages until the message is forwarded.

The Long delay can occur due to the storing and forwarding facility provided by the message switching technique.



Packet Switching

The packet switching is a switching technique in which the message is sent in one go, but it is divided into smaller pieces, and they are sent individually.

The message splits into smaller pieces known as packets and packets are given a unique number to identify their order at the receiving end.

Every packet contains some information in its headers such as source address, destination address and sequence number.

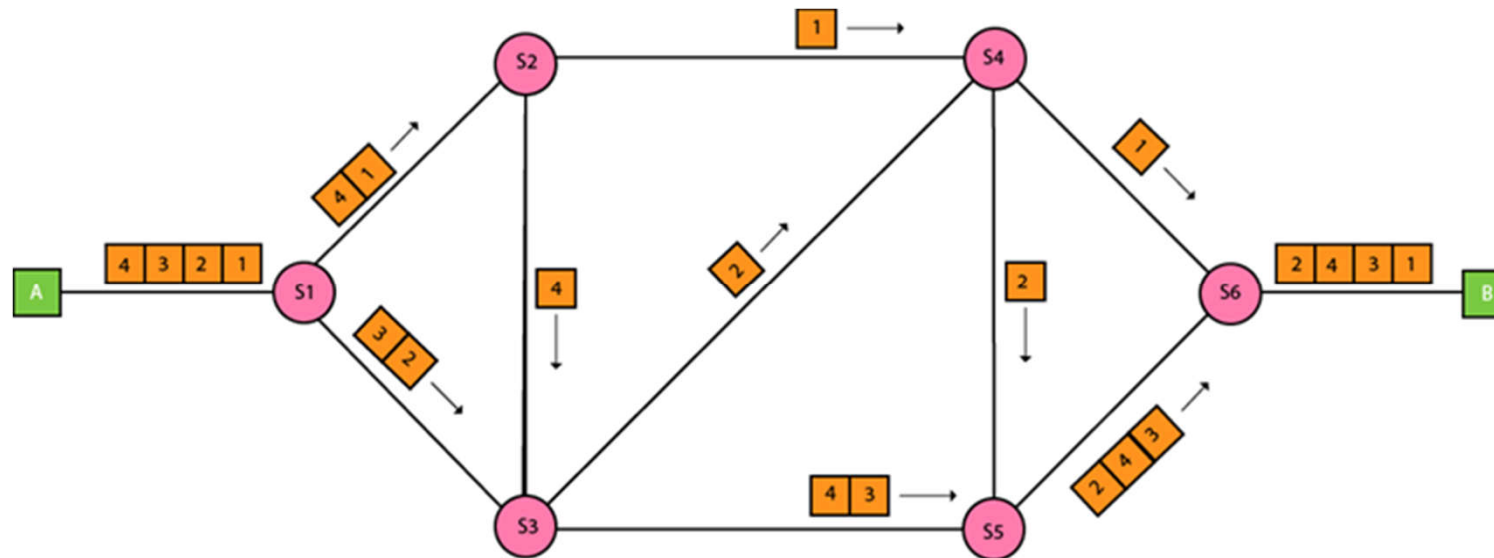
In Packet switching, each data unit just know the final destination address intermediate path is decided by the routers.

Packets will travel across the network, taking the shortest path as possible.

All the packets are reassembled at the receiving end in correct order.

If any packet is missing or corrupted, then the message will be sent to resend the message.

If the correct order of the packets is reached, then the acknowledgment message will be sent.



Approaches Of Packet Switching:

There are two approaches to Packet Switching:

Datagram Packet switching:

It is a packet switching technology in which packet is known as a datagram, is considered as an independent entity. Each packet contains the information about the destination and switch uses this information to forward the packet to the correct destination.

The packets are reassembled at the receiving end in correct order.

In Datagram Packet Switching technique, the path is not fixed.

Intermediate nodes take the routing decisions to forward the packets.

Datagram Packet Switching is also known as connectionless **switching**.

Virtual Circuit Switching

Virtual Circuit Switching is also known as connection-oriented switching.

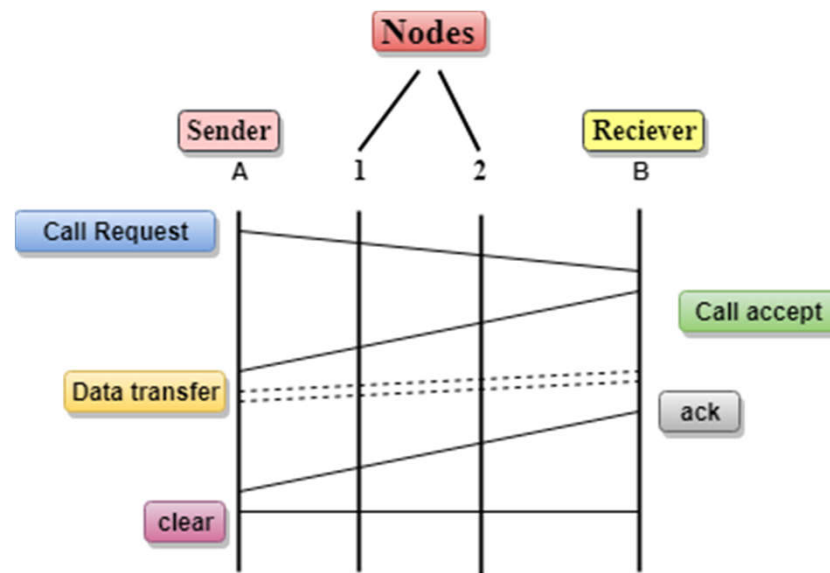
In the case of Virtual circuit switching, a preplanned route is established before the messages are sent.

Call request and call accept packets are used to establish the connection between sender and receiver.

In this case, the path is fixed for the duration of a logical connection.



Let's understand the concept of virtual circuit switching through a diagram:



In the above diagram, A and B are the sender and receiver respectively. 1 and 2 are the nodes.

Call request and call accept packets are used to establish a connection between the sender and receiver.

When a route is established, data will be transferred.

After transmission of data, an acknowledgment signal is sent by the receiver that the message has been received.

If the user wants to terminate the connection, a clear signal is sent for the termination.



Differences b/w Datagram approach and Virtual Circuit approach

Datagram approach	Virtual Circuit approach
Node takes routing decisions to forward the packets.	Node does not take any routing decision.
Congestion cannot occur as all the packets travel in different directions.	Congestion can occur when the node is busy, and it does not allow other packets to pass through.
It is more flexible as all the packets are treated as an independent entity.	It is not very flexible.

Advantages Of Packet Switching:

Cost-effective: In packet switching technique, switching devices do not require massive secondary storage to store the packets, so cost is minimized to some extent. Therefore, we can say that the packet switching technique is a cost-effective technique.

Reliable: If any node is busy, then the packets can be rerouted. This ensures that the Packet Switching technique provides reliable communication.

Efficient: Packet Switching is an efficient technique. It does not require any established path prior to the transmission, and many users can use the same communication channel simultaneously, hence makes use of available bandwidth very efficiently.

Disadvantages Of Packet Switching:

Packet Switching technique cannot be implemented in those applications that require low delay and high-quality services.

The protocols used in a packet switching technique are very complex and requires high implementation cost.

If the network is overloaded or corrupted, then it requires retransmission of lost packets. It can also lead to the loss of critical information if errors are not recovered.

Circuit Switching

In circuit switching there are 3 phases:

- i) Connection Establishment.
- ii) Data Transfer.
- iii) Connection Released.

In circuit switching, each data unit know the entire path address which is provided by the source.

In Circuit switching, data is processed at source system only

Delay between data units in circuit switching is uniform.

Resource reservation is the feature of circuit switching because path is fixed for data transmission.

Circuit switching is more reliable.

Wastage of resources are more in Circuit Switching

It is not a store and forward technique.

Transmission of the data is done by the source.

Congestion can occur during the connection establishment phase, because there might be a case where a request is being made for a channel but the channel is already occupied.

Circuit switching is not convenient for handling bilateral traffic.

In Circuit switching, charge depend on time and distance, not on traffic in the network.

Recording of packet is never possible in circuit switching.

In Circuit Switching there is a physical path between the source and the destination

Circuit Switching does not support store and forward transmission

Packet Switching

In Packet switching directly data transfer takes place.

In Packet switching, each data unit just know the final destination address intermediate path is decided by the routers.

In Packet switching, data is processed at all intermediate node including source system.

Delay between data units in packet switching is not uniform.

There is no resource reservation because bandwidth is shared among users.

Packet switching is less reliable.

Less wastage of resources as compared to Circuit Switching

It is a store and forward technique.

Transmission of the data is done not only by the source, but also by the intermediate routers.

Congestion can occur during data transfer phase, large number of packets comes in no time.

Packet switching is suitable for handling bilateral traffic.

In Packet switching, charge is based on the number of bytes and connection time.

Recording of packet is possible in packet switching.

In Packet Switching there is no physical path between the source and the destination

Packet Switching supports store and forward transmission

Spanning Tree Protocol (**STP**) is a Layer 2 protocol that runs on bridges and **switches**. The specification for **STP** is IEEE 802.1D. The main purpose of **STP** is to ensure that you do not create loops when you have redundant paths in your network. Loops are deadly to a network.11-Dec-2019